



Loumi TREMAS

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PROFESSIONAL SUMMARY

Research Engineer and PhD candidate (defence expected 2026) specialising in **supervised deep learning for physical simulation** and generative models for inverse design. Full-stack research ownership: literature review, FDTD simulation pipeline, model training, benchmarking and deployment – end-to-end. Self-driven learner expanding into **self-supervised learning** (JEPA, Transformers, ViT) through hands-on reimplementations – see portfolio. Proven $10^6\times$ inference speedup over classical solvers at STMicroelectronics.

TECHNICAL SKILLS

ML & DL: PyTorch, JAX/Flax, CNNs (U-Net, ResNeXt), Generative Models, Self-Supervised Learning (JEPA – from scratch), Transformers, ViT, HuggingFace.

Infrastructure: Python, CUDA, Distributed Training (Ray, multi-GPU), Azure, Linux, C.

Research: FDTD Simulation, Inverse Design, Surrogate Modelling, Benchmarking. **Languages:** French (Native), English (Professional)

PROFESSIONAL EXPERIENCE

PhD Researcher – Deep Learning for Physical Simulation (CIFRE)

2023 – 2026 (expected)

Paris-Saclay University $\times$ STMicroelectronics

Institut d'Optique Graduate School, Grenoble

- **Full-stack research ownership:** literature review, dataset design, FDTD simulation pipeline, model training, benchmarking and deployment – independently end-to-end.
- Trained supervised deep learning models (U-Net, ResNeXt) on electromagnetic simulation data – $10^6\times$ inference acceleration over classical FDTD solvers.
- Designed inverse design workflows combining gradient-based optimisation for photonic metasurface design. Published at SPIE, SISPAD and META.
- Orchestrated distributed training on GPU clusters (Ray on Azure); end-to-end profiling and deployment.

AI Engineer

Aug 2022 – Feb 2023

STMicroelectronics

Crolles, France

- Built computer vision pipelines for automated anomaly detection in SEM images; hardened prototypes into production tools. Retained as CIFRE PhD researcher.

EDUCATION

PhD in AI & Computational Physics

2023 – 2026 (expected)

Paris-Saclay University $\times$ STMicroelectronics

Institut d'Optique Graduate School

MSc Artificial Intelligence

2022

Grenoble INP – Ensimag

Grenoble, France

BSc Computer Science

2017 – 2021

Aix-Marseille University (Erasmus: Uppsala Universitet, Sweden)

France

ML PORTFOLIO – PERSONAL PROJECTS

I-JEPA – from scratch

PyTorch · SSL · RTX 3090

- Reimplementation of *Image JEPA* (Assran et al., CVPR 2023): patch masking, dual-encoder EMA ($\tau=0.996$), predictor Transformer. STL-10 (100k): loss $0.408\rightarrow0.088$ (–78%), 100 epochs, ~72 min.

LeWorldModel – from scratch (arXiv:2603.19312)

World Models · Robotics · AdaLN · SIGReg

- Single ViT encoder, AdaLN action conditioning, SIGReg anti-collapse (no EMA), causal temporal masking. Dataset: facebook/jeпа-wms pusht (224 $\times$ 224 RGB, 2-dim continuous actions).

PUBLICATIONS & CONFERENCES

- Diffusion Schrödinger Bridges for Metasurface Inverse Design arXiv:2602.06605, 2026
- Enhanced Posterior Sampling via Diffusion Models for Metasurface Inverse Design arXiv:2601.15210, 2026
- Heuristic-Initialized Gradient Descent for Large-Scale Metasurface Inverse Design SPIE, 2026
- Inverse Design of Optical Metasurface for CMOS Imagers: Multi-Objective Approach SISPAD, 2025
- Transition to Neural Network Solutions for Flat Optical Devices SISPAD, 2024
- Self-Enhanced Generative Metasurface Inverse Design META, Dublin 2026
- Impact of Numerical Simulation Boundary-Decomposition Errors META, Toyama 2024

INTERESTS & HOBBIES

Endurance Sports: Running and triathlon. **Automotive:** 4th place at the 4L Trophy (1 000+ teams) – 6 000 km humanitarian rally across the Moroccan desert.